

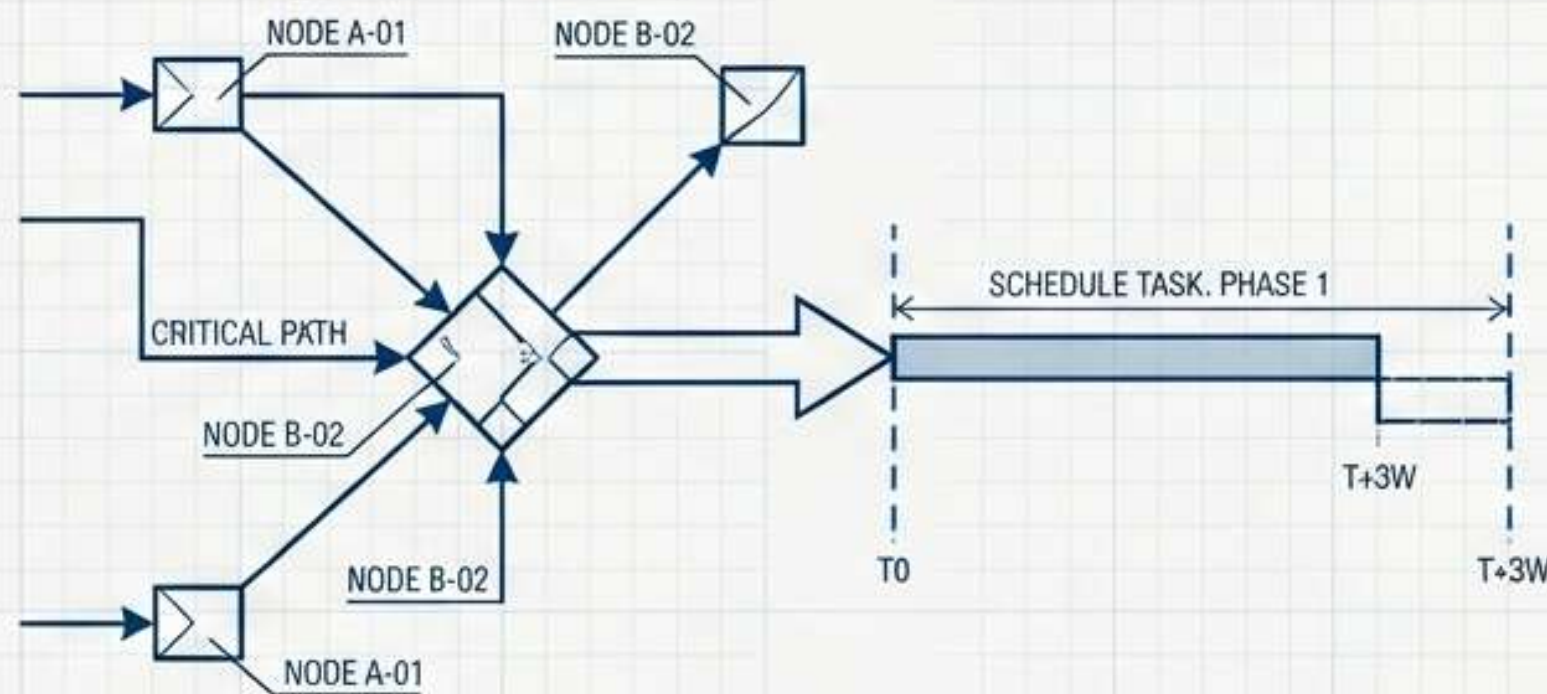


ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG QUỐC TẾ
VNU-INTERNATIONAL SCHOOL

BASIC PROJECT PLANNING TECHNIQUES & SCHEDULE PLANNING

INS3044: IT Project Management – Topic 4

Lecturer: Dr. NGUYEN The Loc



Basic Project Planning Techniques



Understand the planning purpose.



Identify six planning questions.



Define Project Execution Plan.



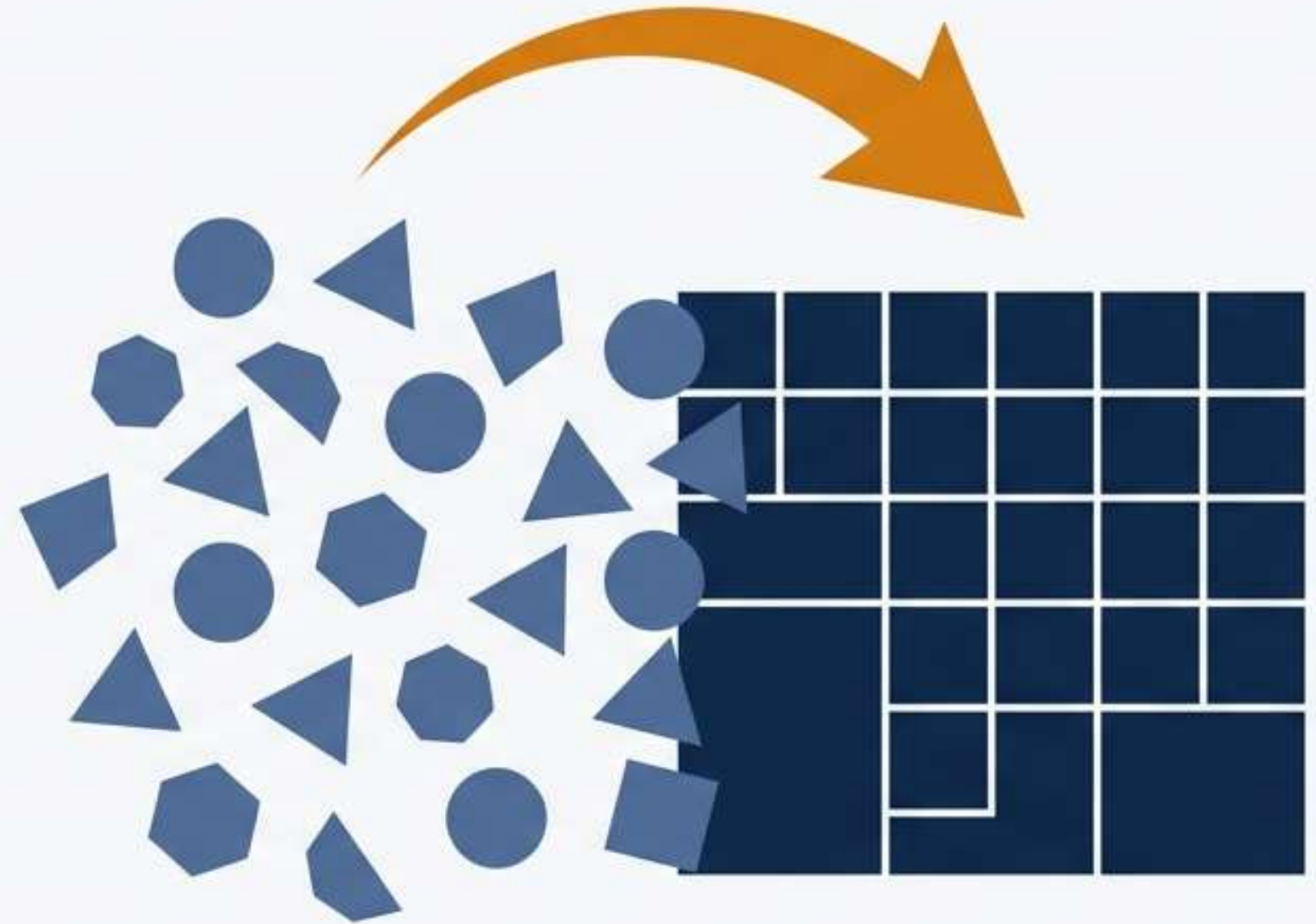
Create Work Breakdown Structures.



Schedule project events.

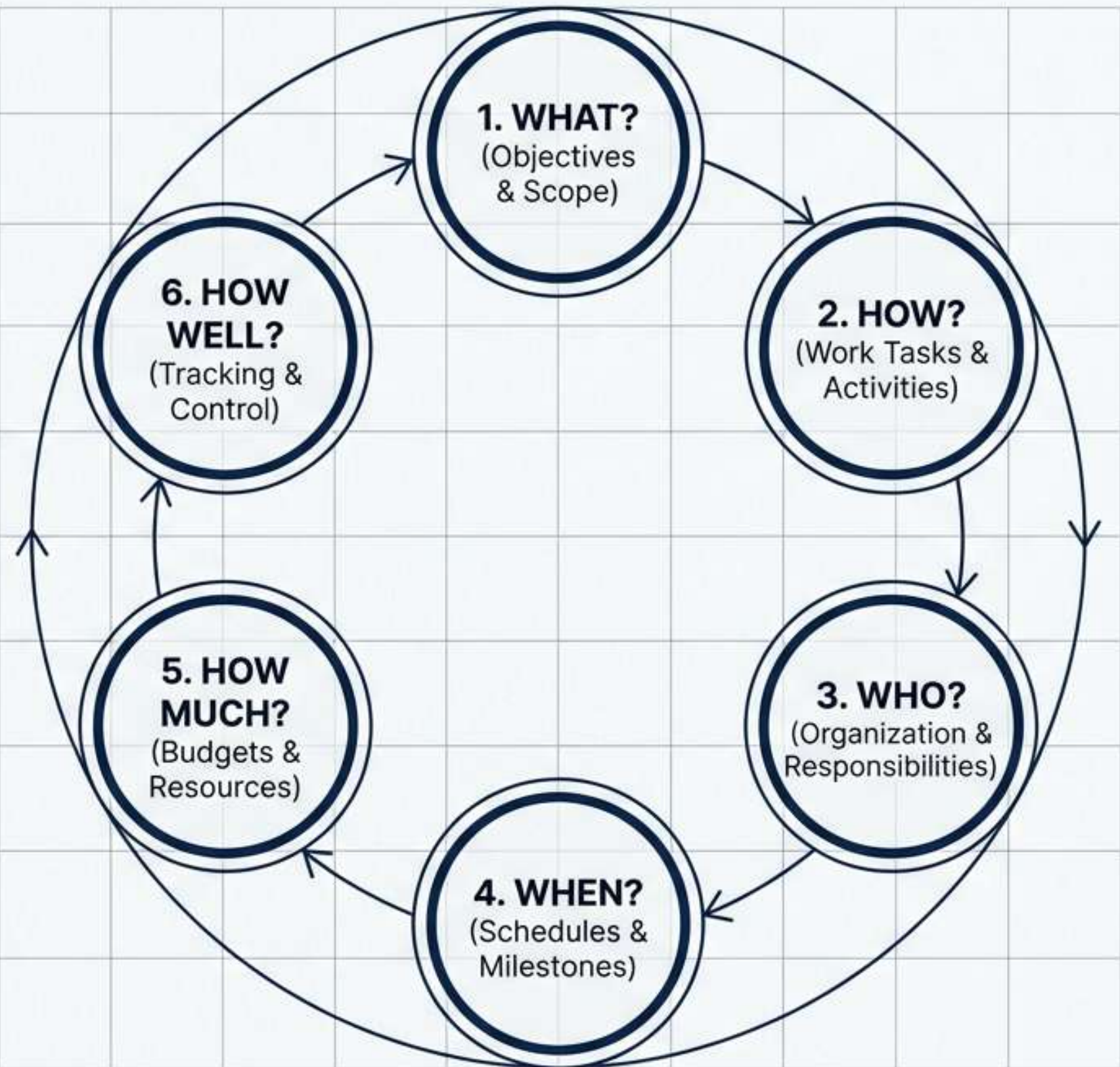
Why Do We Plan?

- Projects are inherently unique.
- Uniqueness brings high uncertainty.
- Planning answers basic questions.
- Planning precedes actual execution.
- Goals dictate the plan.



The Core of Project Planning

Every unique project poses new uncertainties. Planning is the systematic process of answering six fundamental questions before work begins:



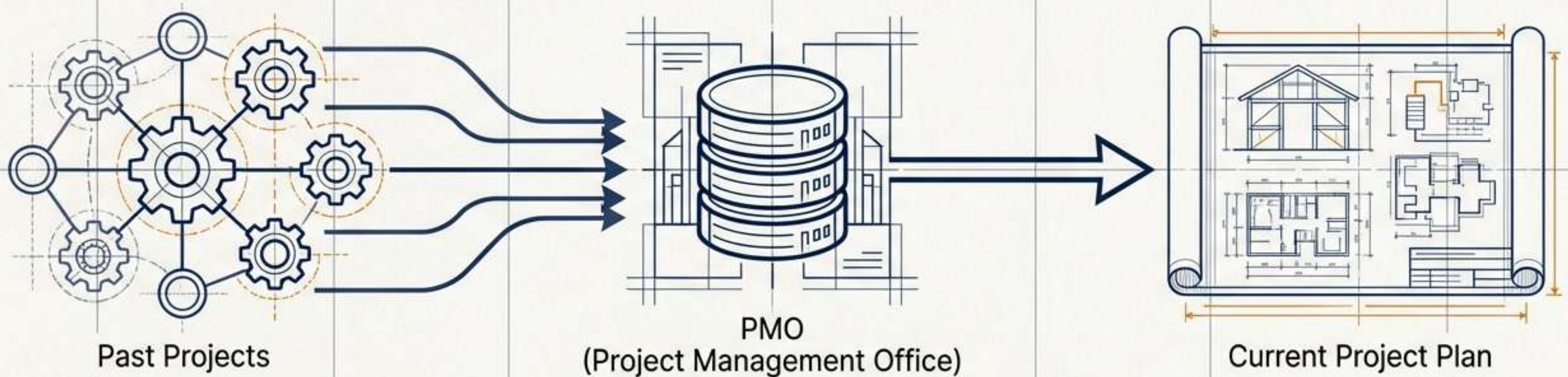
The Blueprint: The Project Execution Plan

The Execution Plan houses the answers to all six questions. It includes the Statement of Work, Work Breakdown Structure, responsibilities, schedules, budgets, and quality plans.



Step Zero: Learning from the Past

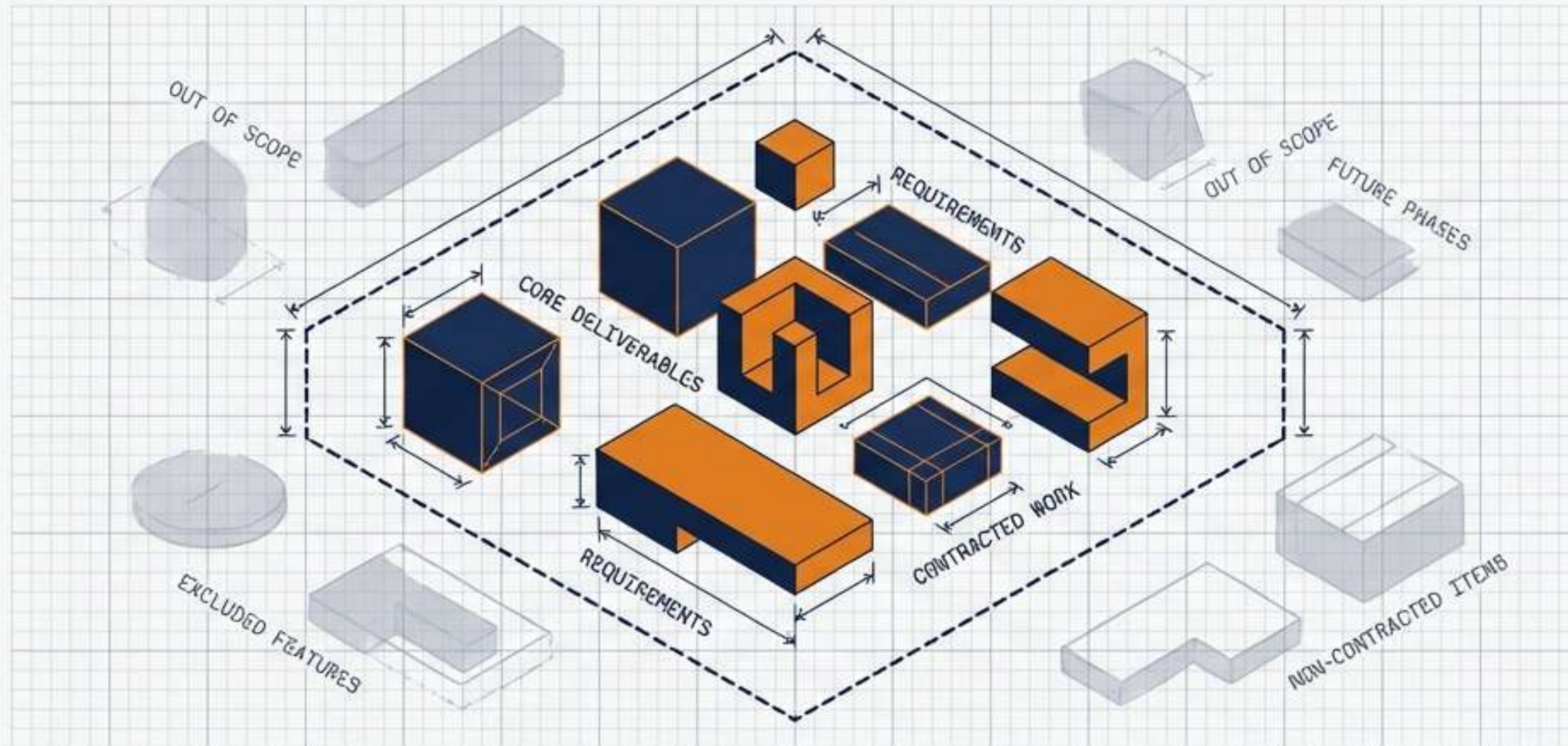
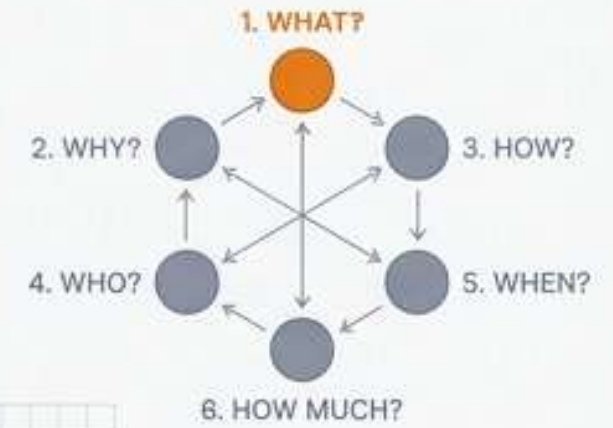
No project is ever totally unique. World-class project managers avoid treating every challenge as a blank slate.



- Consulting the PMO (Project Management Office).
- Reviewing formal postmortems and Lessons Learned.
- Utilizing existing methodologies and templates to avoid reinventing the wheel.

Step 1: Defining the “WHAT”

Setting the project scope determines the overall size and extent of work.

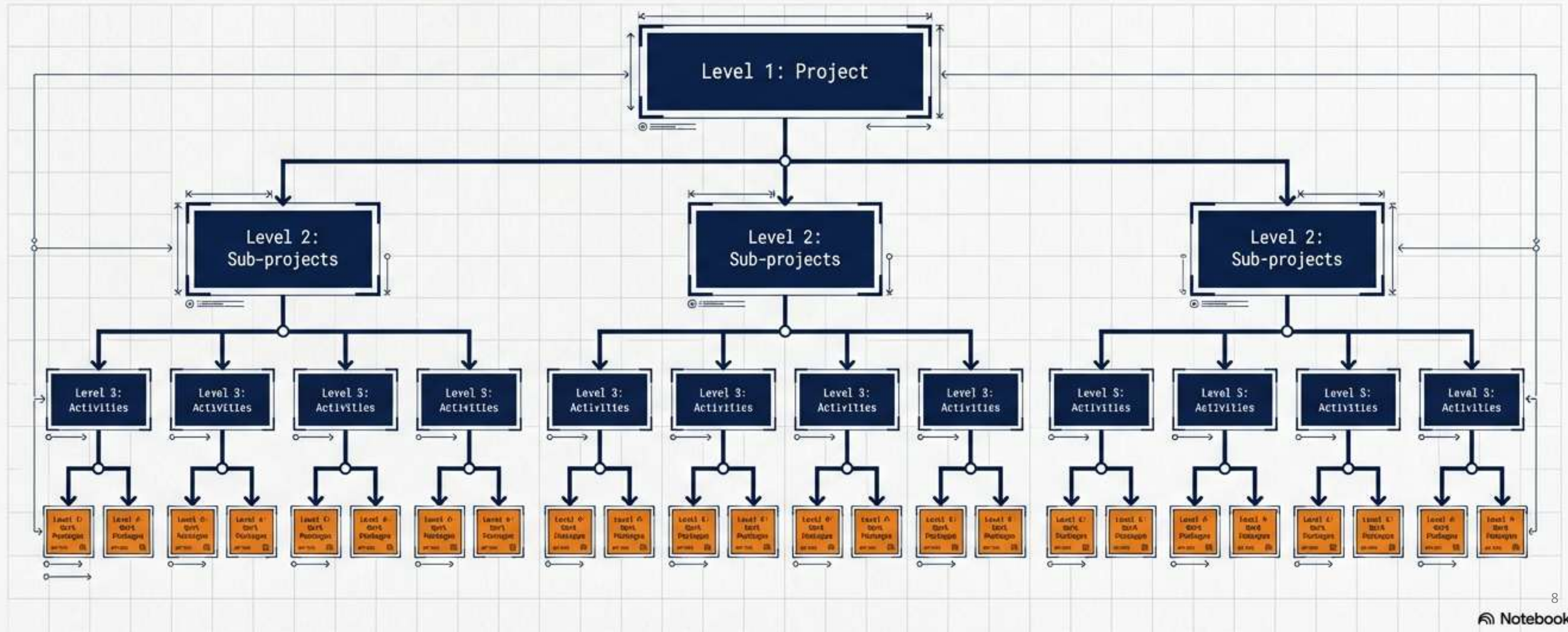
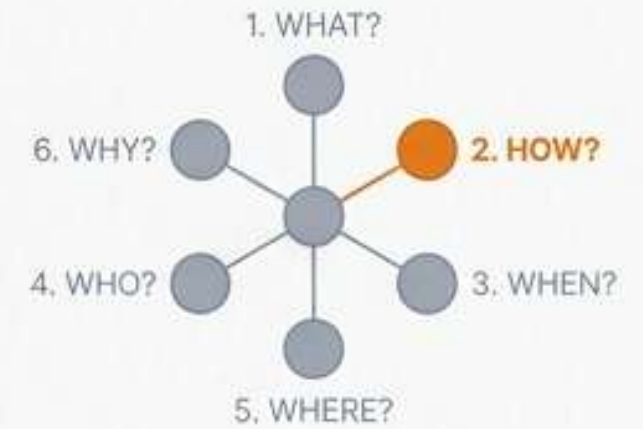


SCOPE: The definitive boundaries of the project, derived from user needs and constraints.

STATEMENT OF WORK (SOW): A formal summary description of the contracted work, deliverables, and end-items.

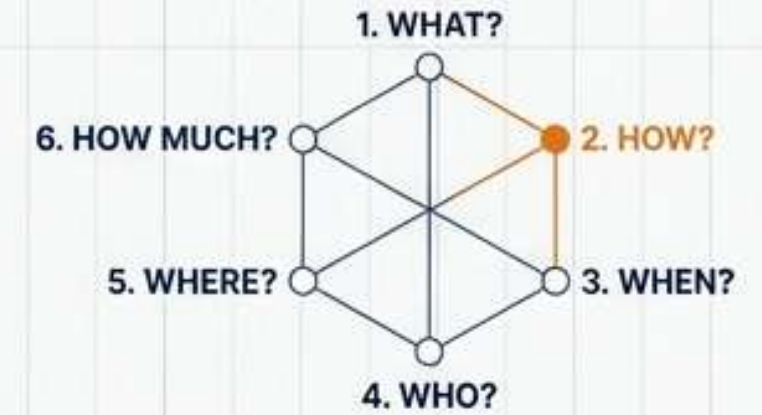
Step 2: Answering “HOW” with the WBS

Complex projects are solved by dividing them into manageable pieces. The Work Breakdown Structure (WBS) systematically reduces the scope into granular elements.



Anatomy of a Work Package

The lowest level of the WBS is the Work Package. It is small enough to be easily planned, budgeted, scheduled, and monitored.



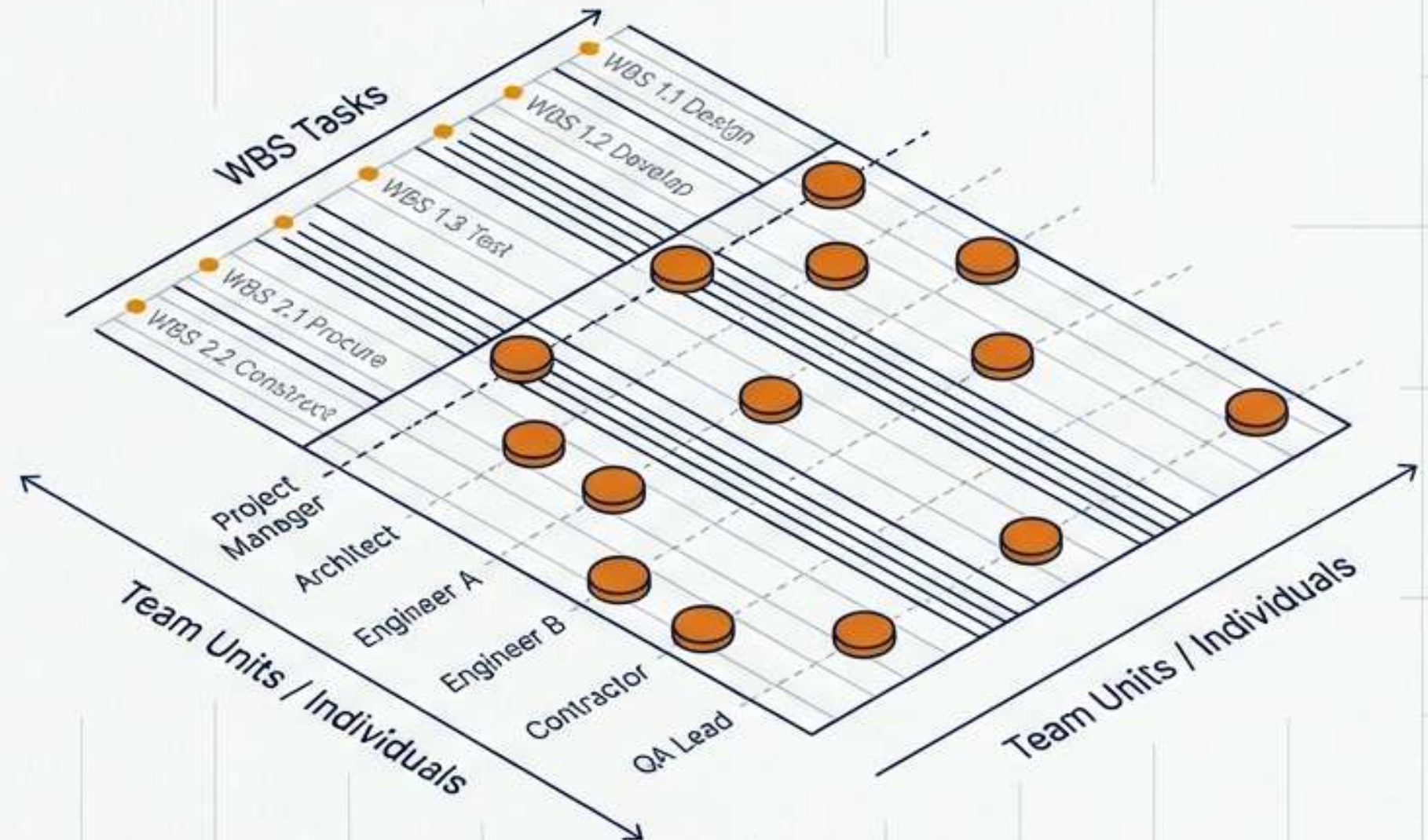
Inputs	Task	Outcomes
Predecessors	Statement of work	Deliverables
Preconditions	Time	Results
Resources	Cost	
Requirements/specifications	Responsibility	
	Quality assurance	
	Risk	

Takeaway Callout: The entire project is ultimately controlled by managing these individual, localized work packages.

Step 3: Assigning the 'WHO'

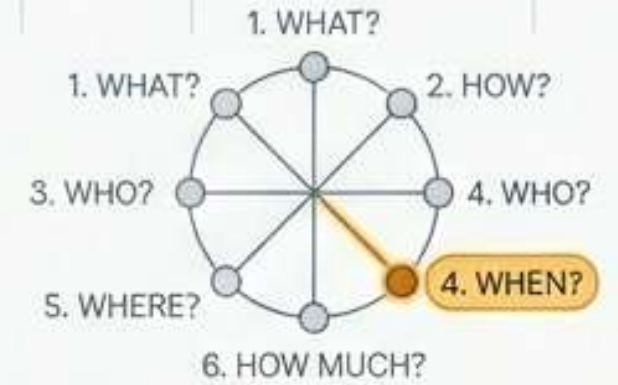
The Responsibility Matrix integrates the project organization with the WBS.

- - Prescribes exact responsibilities for internal units, individuals, and external subcontractors.
- - Ensures strict accountability for every Work Package.
- - Achieves consensus and reduces conflict among participants.



Step 4: The Elements of “WHEN”

Project plans are roadmaps. To read the map, managers must distinguish between three concepts:



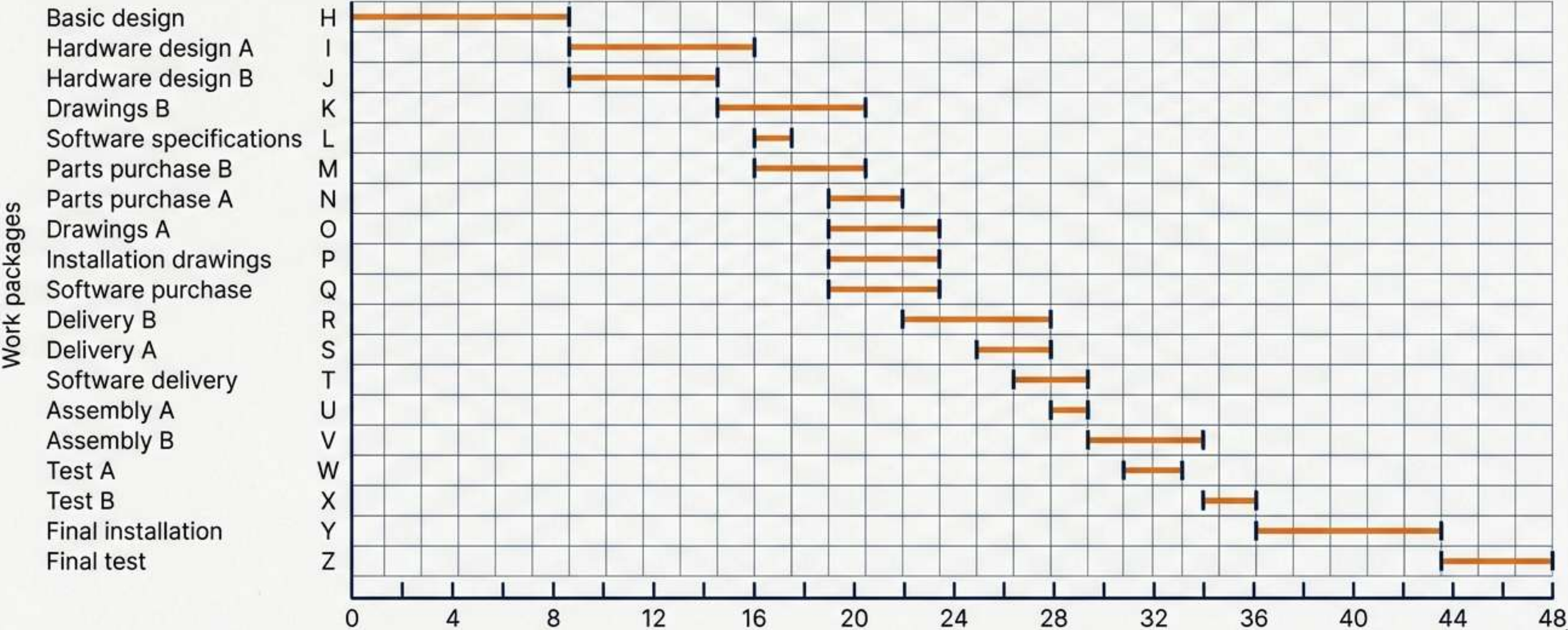
Task/Activity: Consumes time and resources. The actual work being done (the journey).

Event: A specific moment in time. Represents the exact instant a task starts or finishes.

Milestone: A major event or critical signpost showing significant progress along the project roadmap.

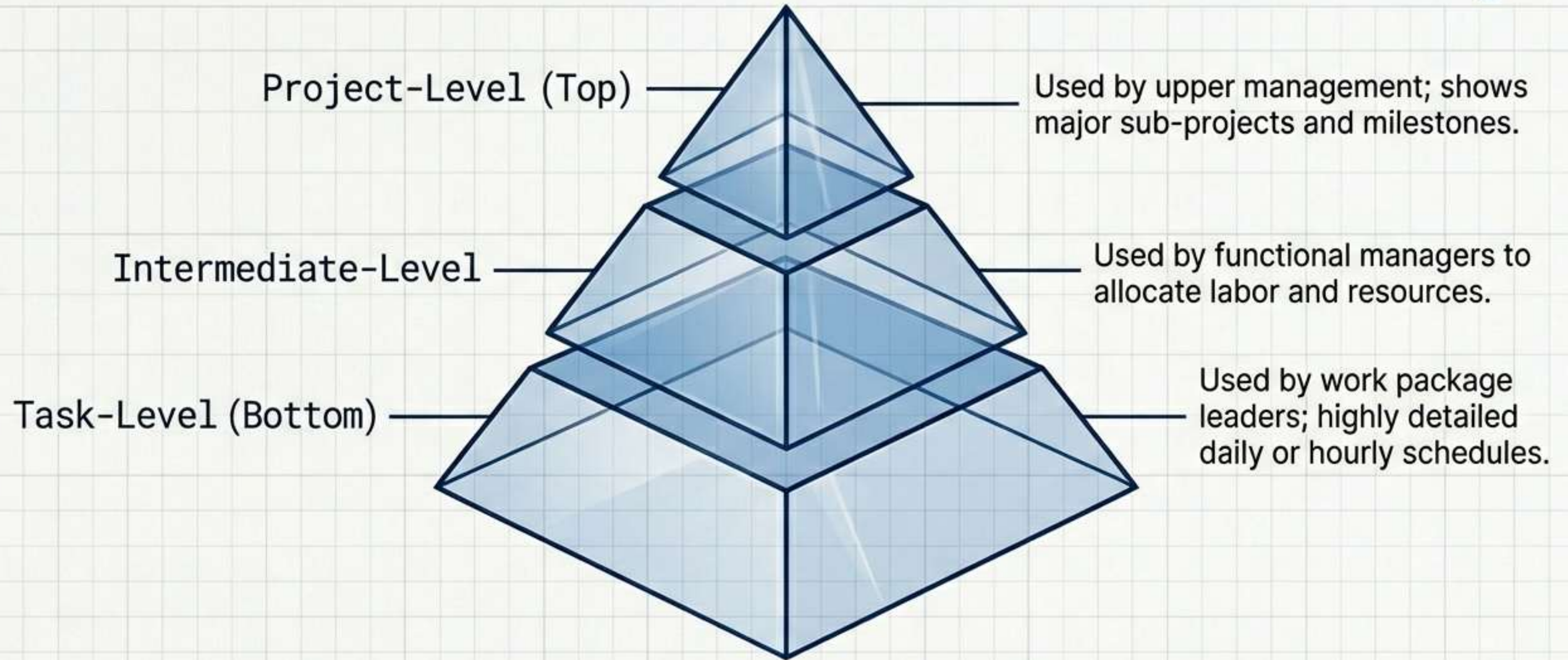
Scheduling Tools: The Gantt Chart

Named after Henry L. Gantt (who developed it for the WWI munitions program), this is the most common visual scheduling technique. Time is the common denominator. Start and completion times of work packages are indicated by the beginning and ending of each bar. Used to easily assess progress vs. time.

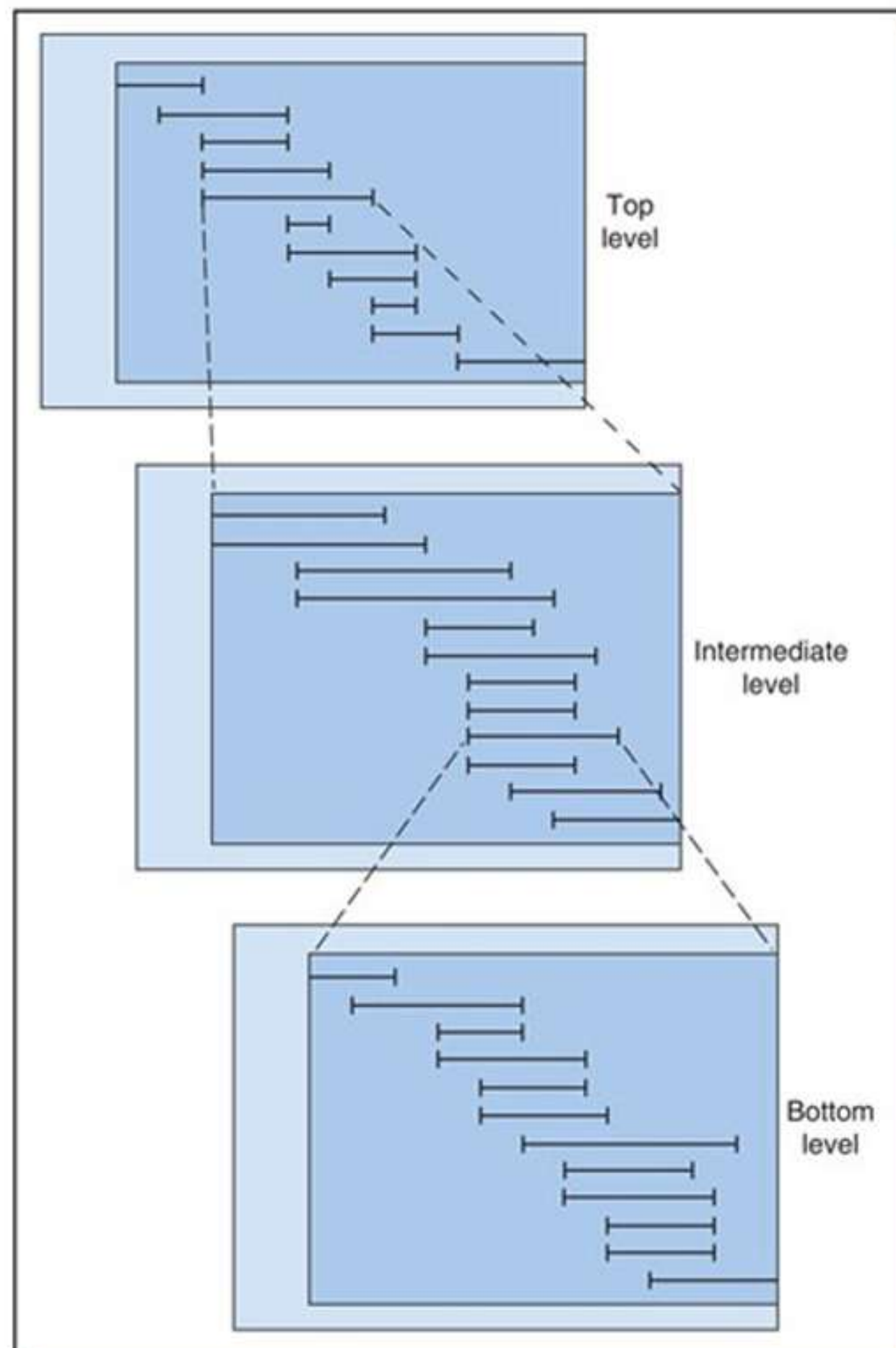


The Hierarchy of Schedules

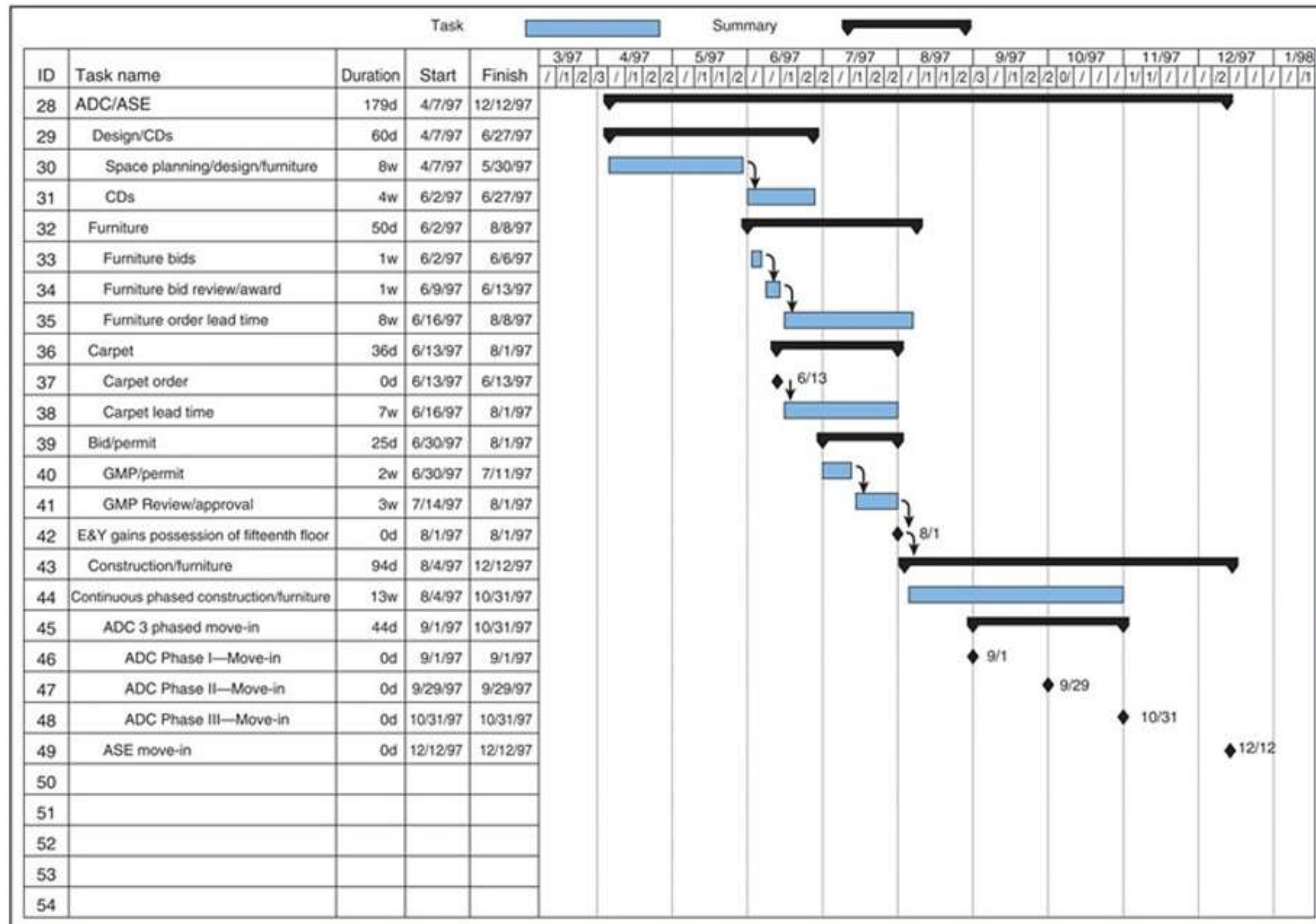
Large projects require schedules that scale to different audiences.



Hierarchy of Schedules

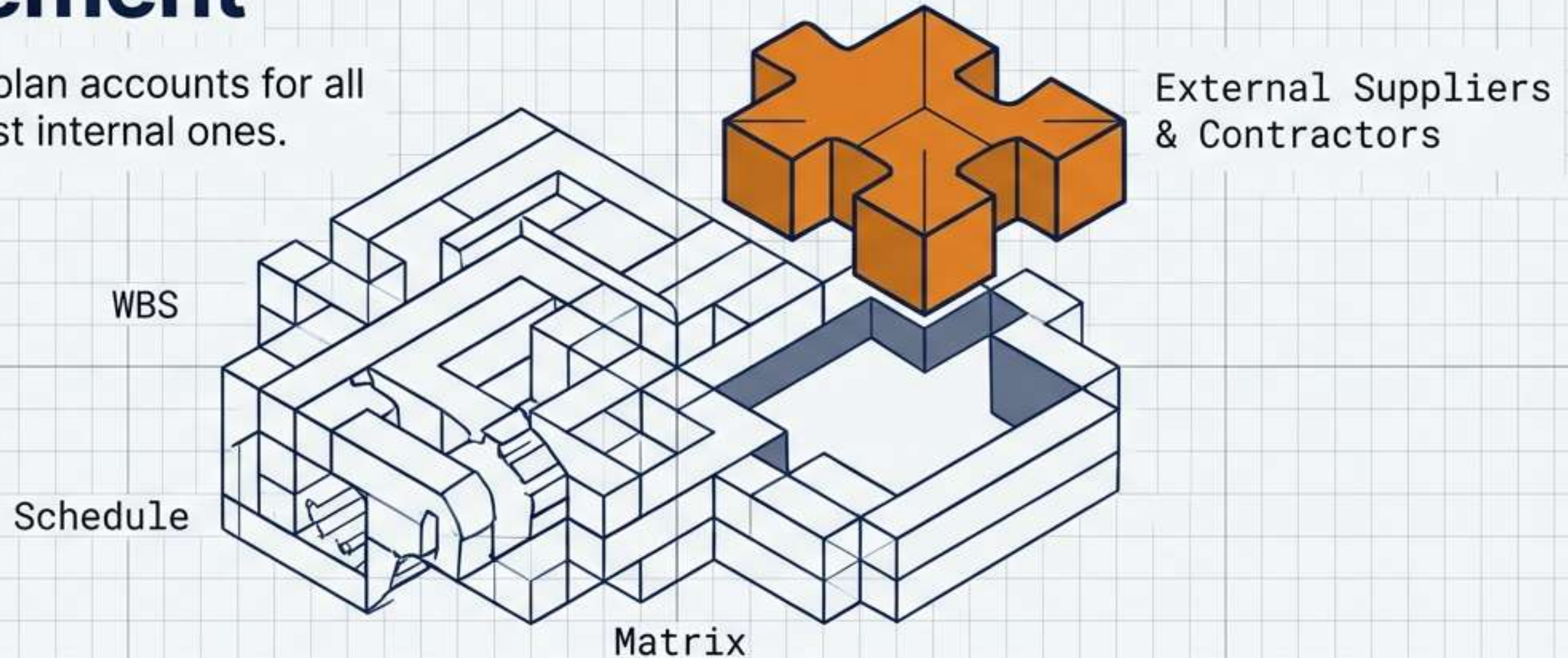


Multi-level Schedule



The Hidden Dimension: Procurement

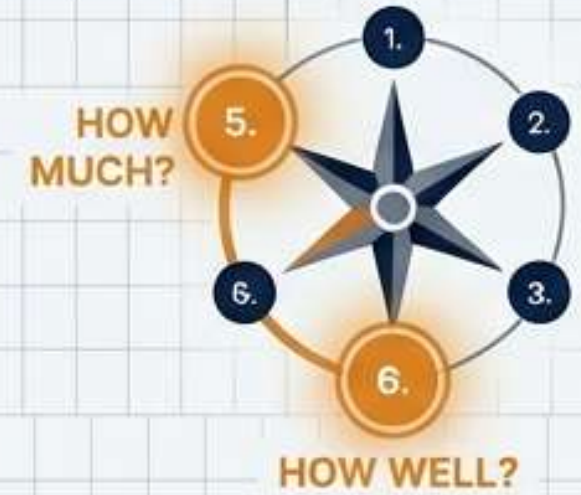
A robust project plan accounts for all resources, not just internal ones.



Integration Requirement: Procured items (external contractors, purchased software, physical supplies) must be explicitly mapped into every element of the plan: the WBS, the schedule, the budget, and the responsibility matrix. They are not afterthoughts.

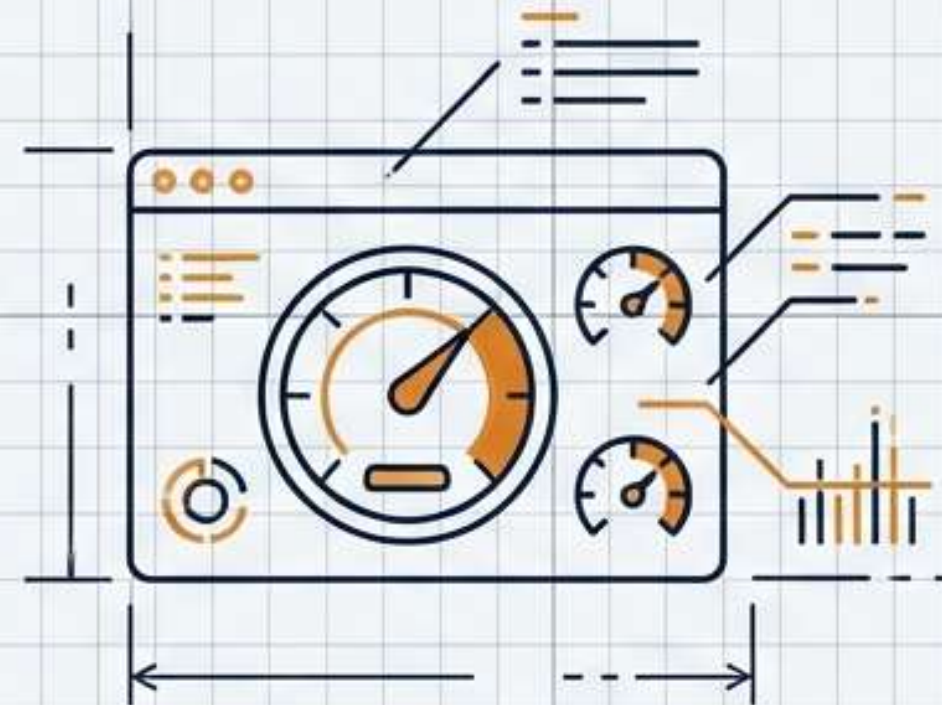
Steps 5 & 6: Budgets and Control

The final steps of the Execution Plan define the financial and qualitative boundaries.



How Much:

Creating budgets and allocating funds directly to specific work packages.

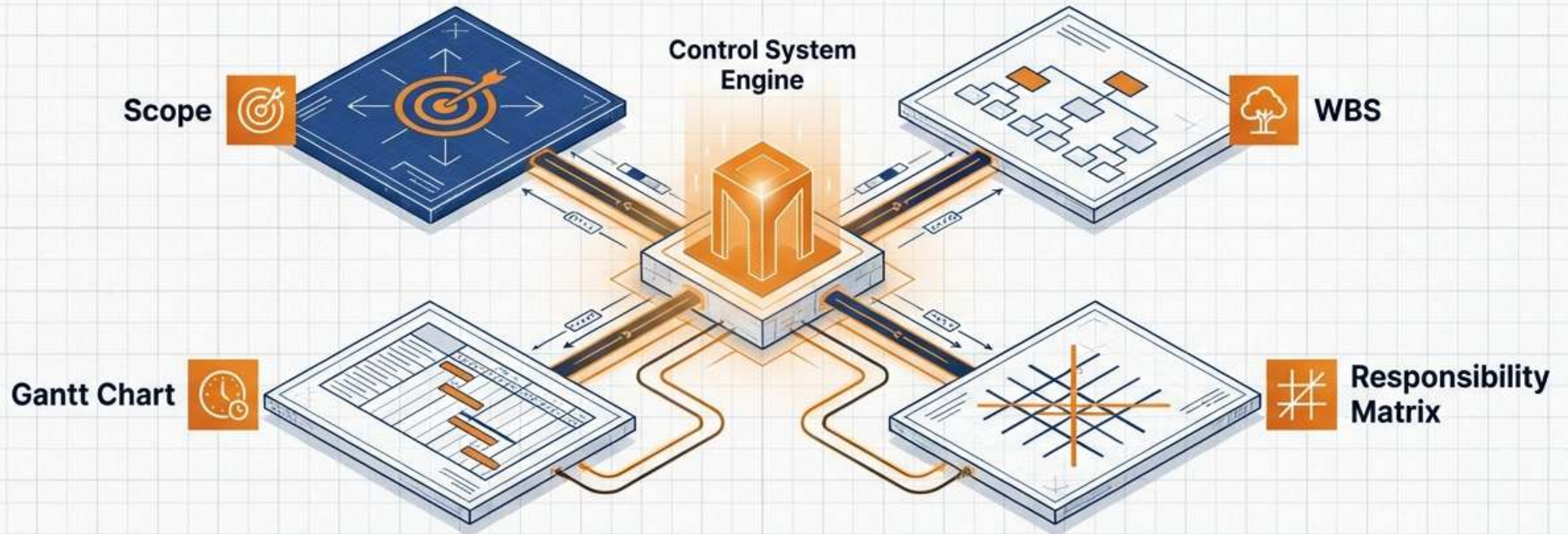


How Well:

Establishing the control framework to track actual progress against the schedule and requirements, enabling corrective action.

The Integrated Systems Approach

Project Planning is not a collection of isolated documents. It is a unified system.



The Scope defines the goal. The WBS breaks it down. The Matrix assigns the people. The Schedule maps the time. Together, they form a single, interrelated control system that allows a project manager to pilot complex endeavors from chaos to completion.

Takeaways & Discussion

- Planning answers six questions.
- PEP guides project execution.
- WBS breaks down work.

1. Why use a WBS?

2. What defines work packages?

3. Why are milestones important?

Project Schedule Planning and Networks

- Understand project network diagrams.
- Apply Activity-on-Arrow methods.
- Identify the critical path.
- Calculate slack and float.
- Convert networks to calendars.
- Implement resource leveling techniques.



The Gantt Chart: Visualizing Time



Pros

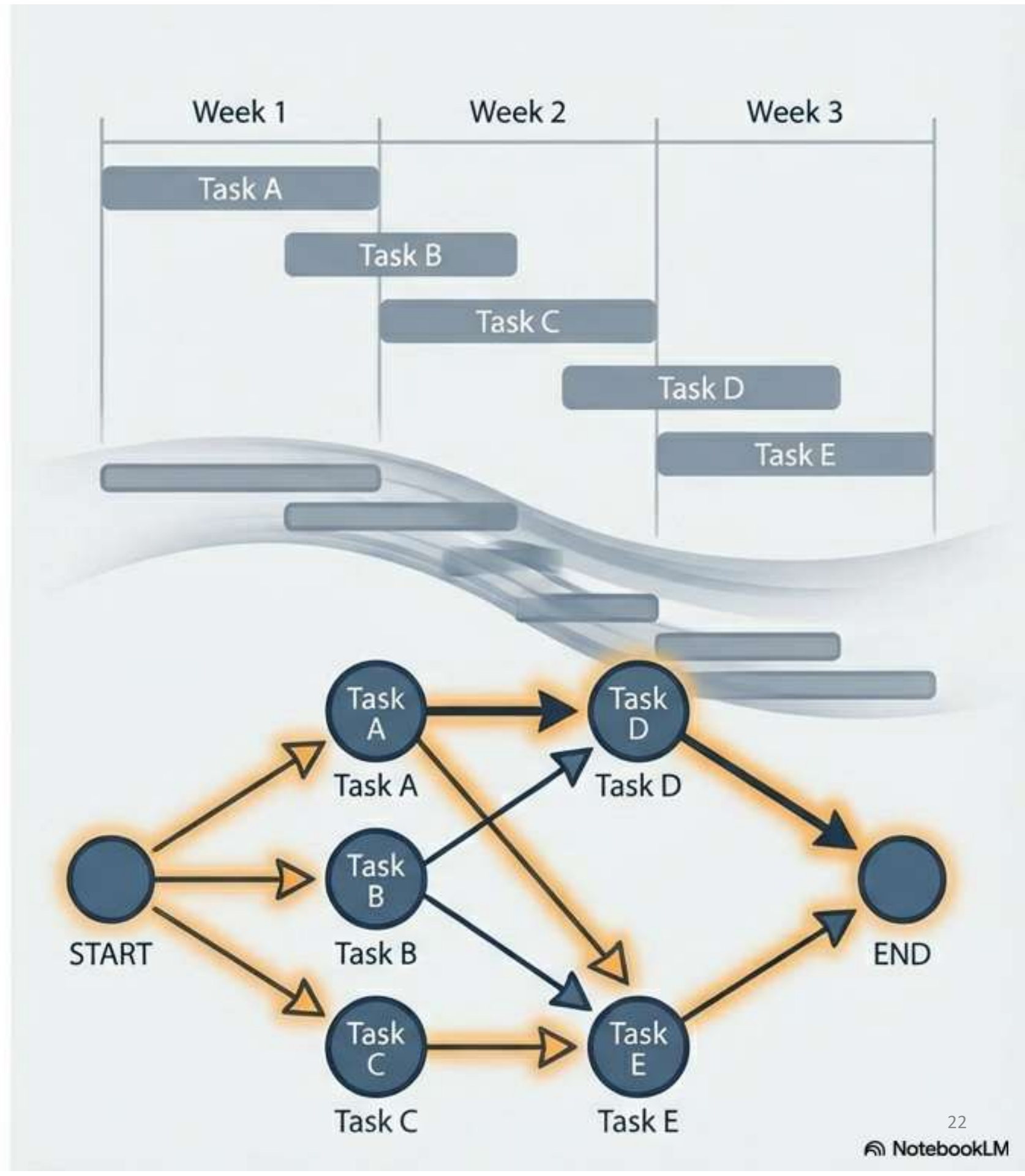
- Standard tool for project communication.
- Visualizes timing against a calendar.
- Effective for tracking progress (Ahead/Behind).

Limitations

- Limitation: Hides complex task dependencies.
- Limitation: Hard to analyze changes.
- Best used AFTER network analysis.

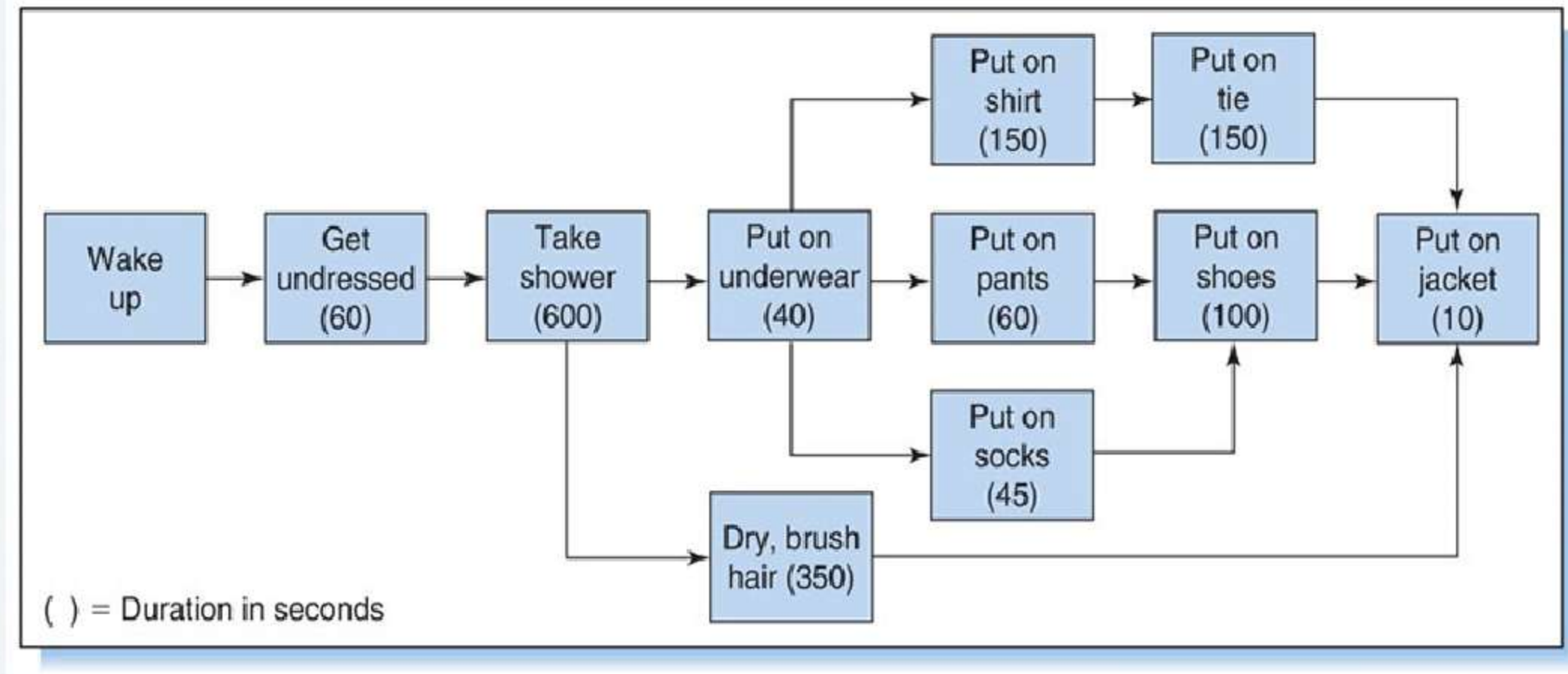
Beyond the Gantt Chart

- Gantt charts show time.
- Networks explicitly show dependencies.
- Networks reveal delay impacts.
- Vital for project control.



Activity-on-Arrow Networks

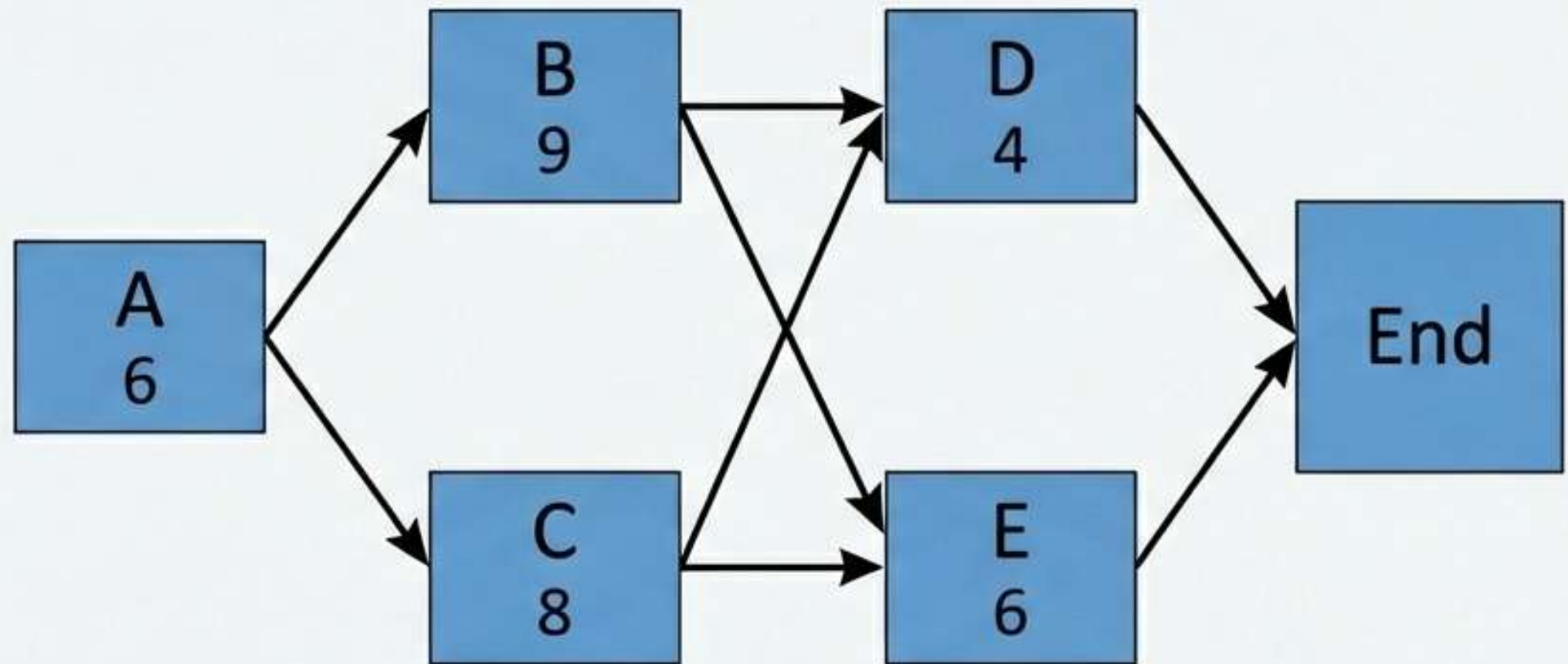
- Nodes represent work tasks.
- Arrows show logical sequence.
- Predecessors must finish first.
- Successors follow completed tasks.



() = Duration in seconds

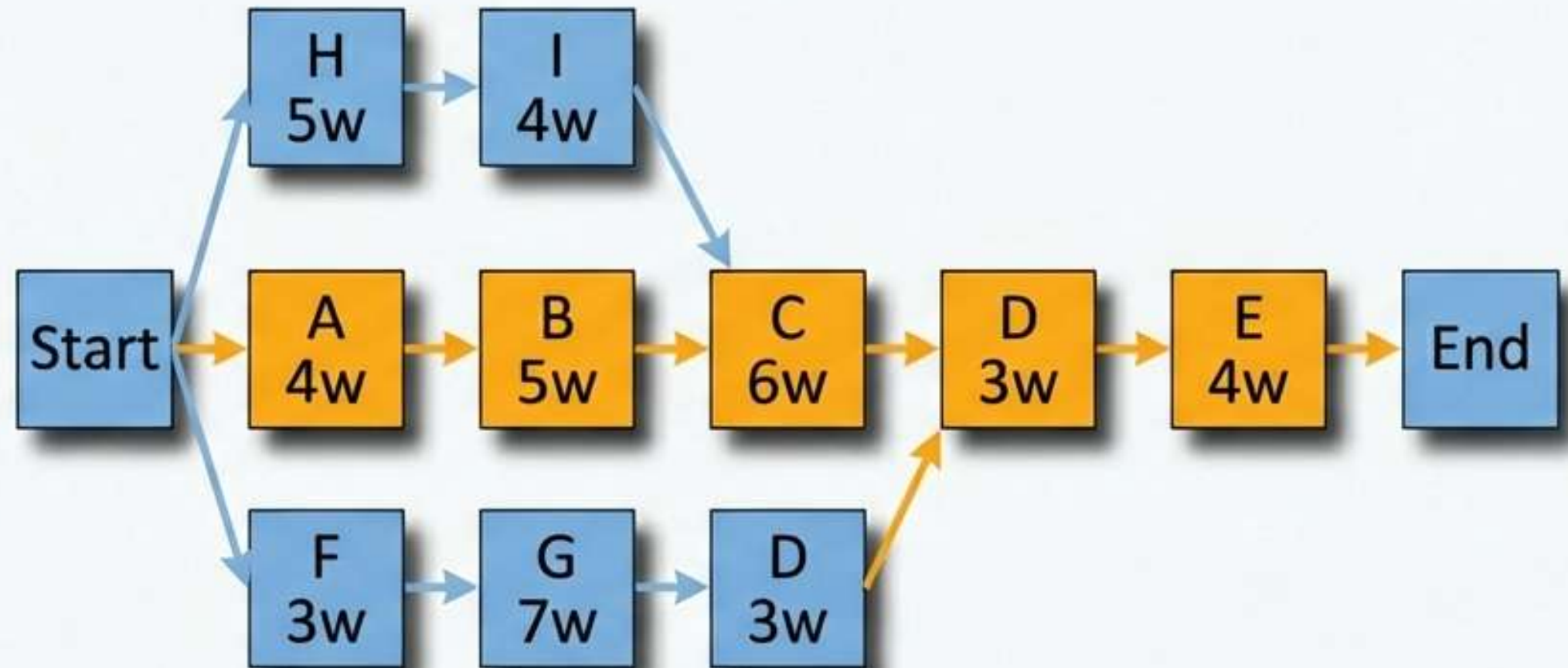
Building the Project Network

- Start with one node.
- Connect immediate predecessors only.
- Avoid redundant dependency lines.
- End with one node.



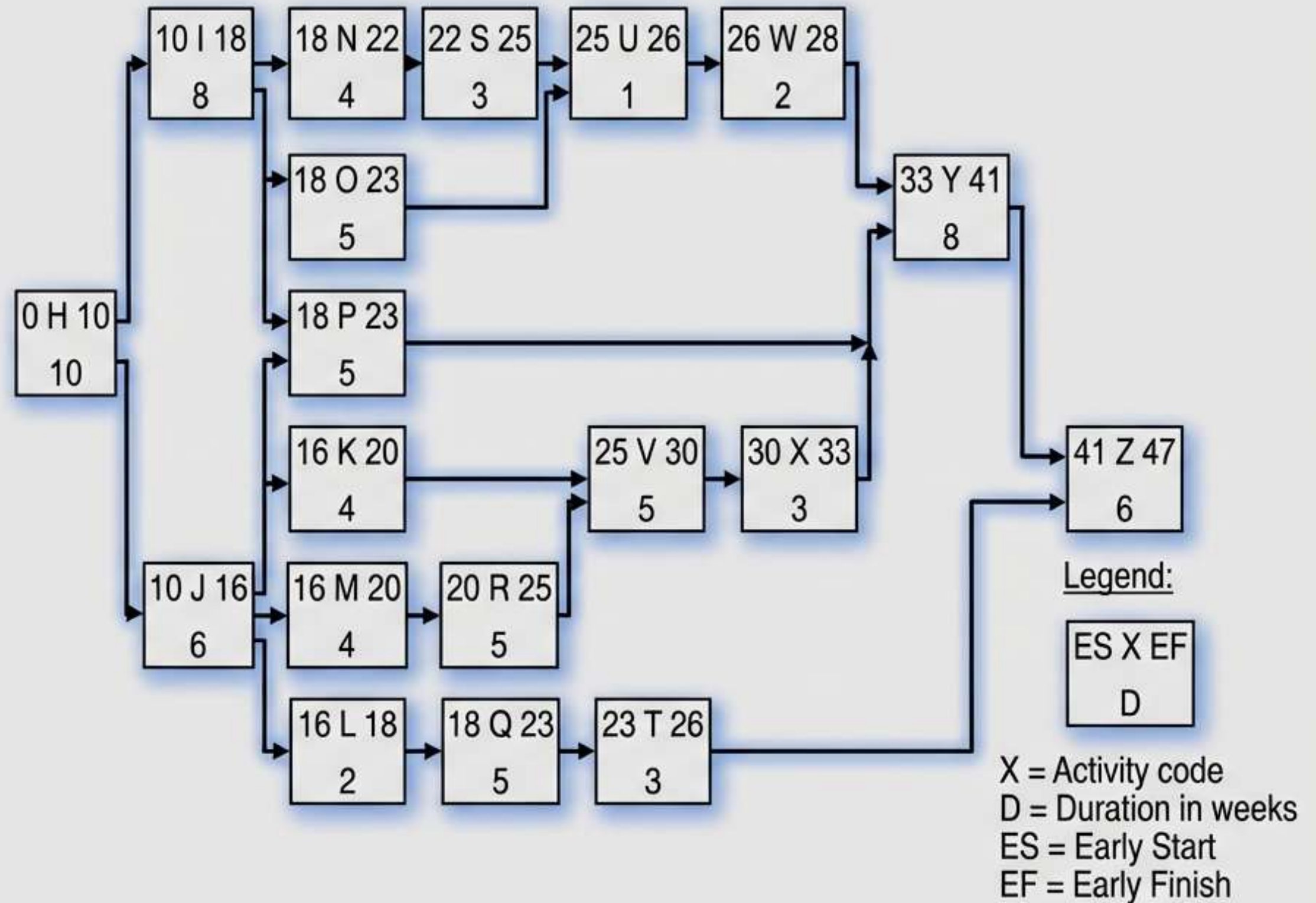
Defining the Critical Path

- Longest path through network.
- Determines shortest project duration.
- Dictates final target deadlines.
- Zero schedule flexibility allowed.



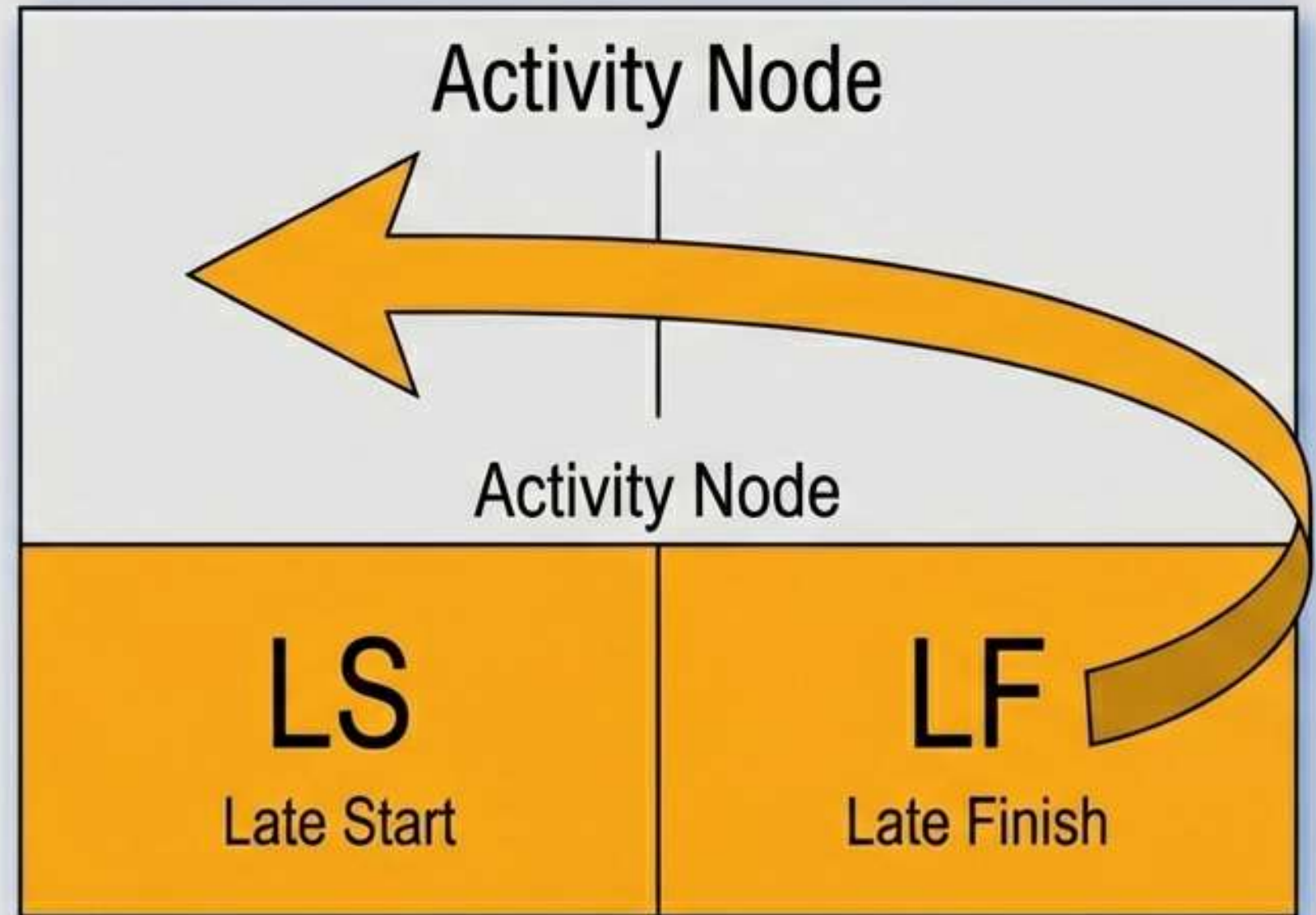
The Forward Pass

- Moves left to right.
- Calculates early start times.
- Calculates early finish times.
- ES plus duration equals EF.



The Backward Pass

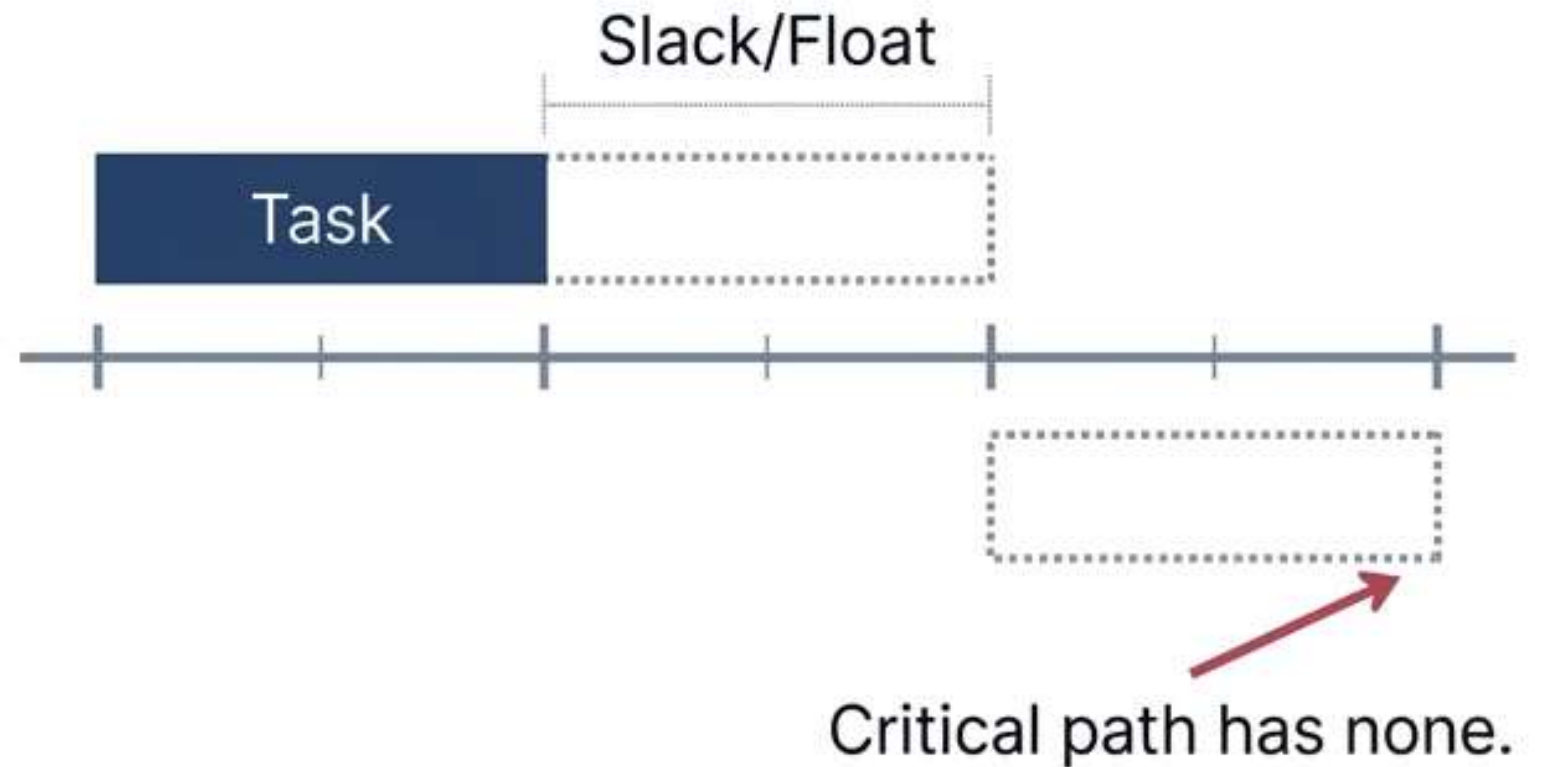
- Moves right to left.
- Calculates late finish times.
- Calculates late start times.
- $LF \text{ minus duration equals } LS$.



Marigold Arrow Indicates Backward Pass

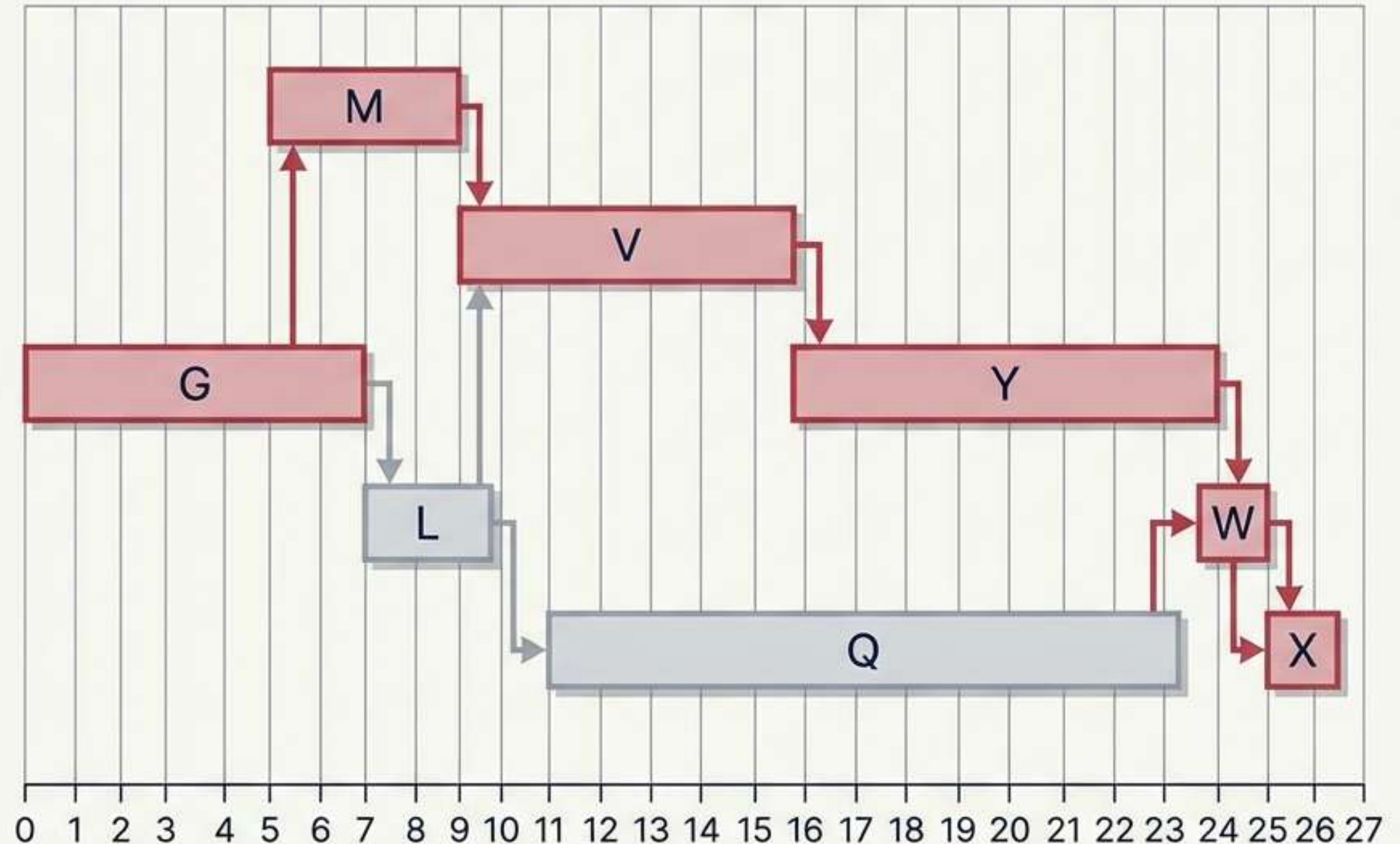
Calculating Slack and Float

- Measures schedule flexibility.
- Total Slack: Overall flexibility.
- Delays without delaying project.
- Free Slack: Task flexibility.
- Delays without delaying successors.
- Critical path has none.



Building Time-Scaled Networks

- Combines network and Gantt.
- Shows explicit task dependencies.
- Visualizes actual duration length.
- Reveals parallel work tracks.
- Highlights schedule slack visually.

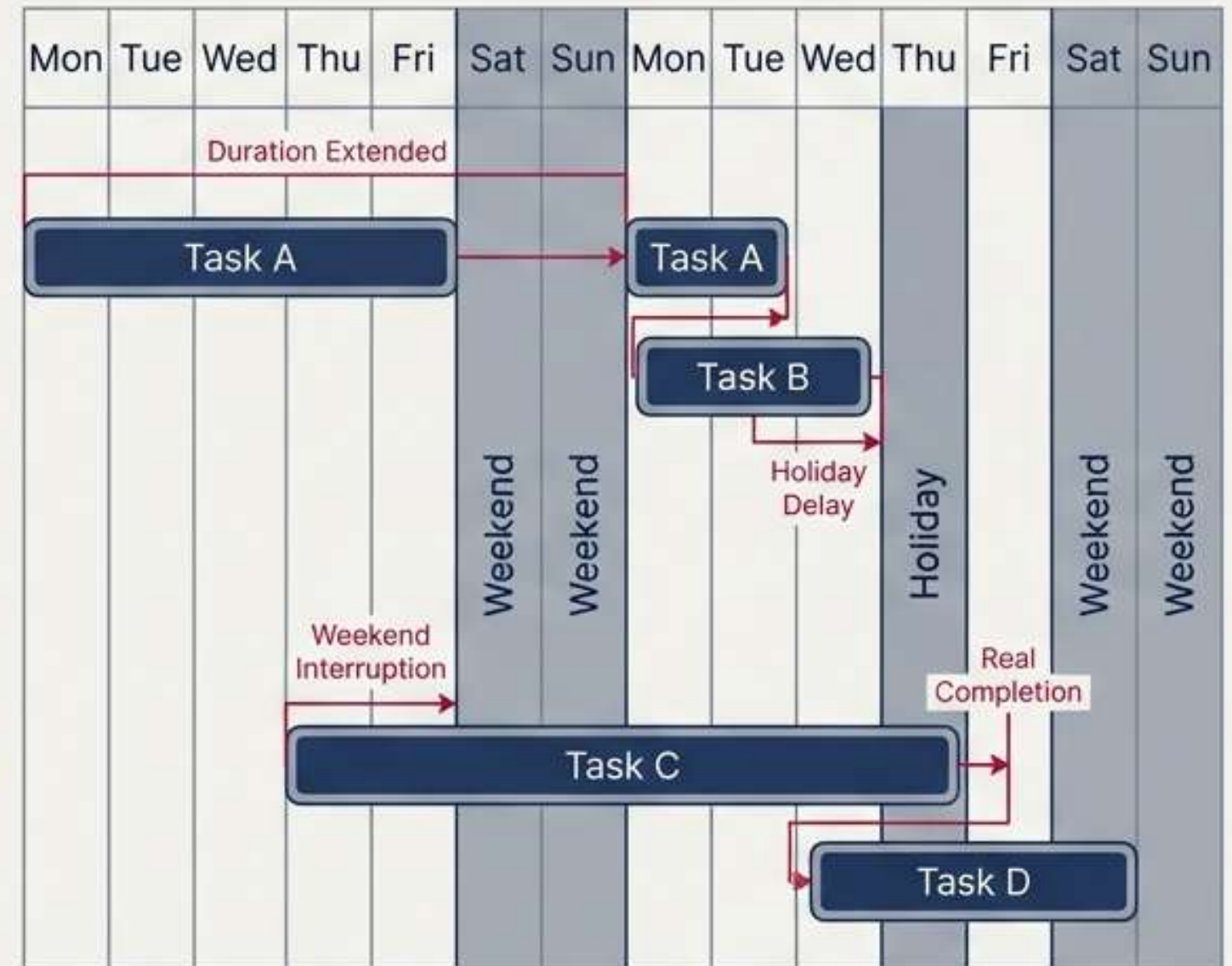


Mapping to Gantt Calendars

- Moves abstract to reality.
- Maps to actual calendars.
- Accounts for strict weekends.
- Accounts for scheduled holidays.
- Tracks true completion dates.
- Facilitates team communication.



Abstract Timeline



Gantt Calendar

Key Takeaways and Practice

Takeaways:

- Networks map project logic.
- Critical paths dictate timelines.
- Resources constrain ideal schedules.

Practice:

Identify your morning critical path.

Where is your schedule float?

How would you level resources?